

A PROJECT REPORT ON

**“ExamAIR – A Digital OMR – Based
Examination Evaluation System”**

Submitted in partial fulfilment of the requirement for the award of

MASTER OF COMPUTER APPLICATIONS

Of

KANNUR UNIVERSITY

By

AMRUTHA BALAN

IT24GMCAD21



SCHOOL OF INFORMATION SCIENCE AND TECHNOLOGY

KANNUR UNIVERSITY CAMPUS

MANGATTUPARAMBA

KANNUR-670567

2024- 2026

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Certificate

*This is to certify that project entitled “ExamAIR” is a bonafide work successfully done by, **AMRUTHA BALAN (IT24GMCAD21)** of Fourth Semester MCA for the partial fulfilment for the award of Master’s Degree from Kannur University during the period 2024-2026.*

Shylaja P

(Internal Guide)

Place: Mangattuparamba

Date:

Dr. Abdul Haleem P P

(Head of the Department)

External Examiners

1)

2)

DECLARATION

I AMRUTHA BALAN , fourth semester MASTER OF COMPUTER APPLICATIONS student, hereby declare that the project report entitled “ ExamAIR - A Digital OMR-Based Examination Evaluation System ” submitted in partial fulfilment of the requirement for the award of the MCA, is a report of the original work done by me during the period of study at School of Information Science and Technology, Mangattuparamba under the supervision of Shylaja P, Internal guide , Department of IT, School of Information Science and Technology, Mangattuparamba.

PLACE: MANGATTUPARAMBA

AMRUTHA BALAN

DATE:

ACKNOWLEDGMENT

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With the showers of blessing of God, I kept fit enough throughout the project for its successful completion. I submit my success and my project work and myself to the Almighty.

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My acknowledgement would not be complete without acknowledging my gratitude to my beloved parents who have been the pillars of support and constant encouragement throughout the course of this project.

AMRUTHA BALAN

ABSTRACT

ExamAIR is a digital tool designed to manage, prepare, and assess OMR (Optical Mark Reading) response sheets. By integrating Web, mobile, and deep learning technologies, it is intended to reduce manual labour, improve evaluation accuracy, and speed up the process of producing results.

The platform is made up of a mobile application created with Flutter and a web-based administrative system created with Python Django. By facilitating communication between the front end and back end, REST APIs provide safe and effective data flow between various modules. The system can be built up to support multiple roles, each with a unique set of features and permissions, including Administrator, Employee, and Evaluator.

Employees can be managed by the administrator, who can also build exams, set up test centers and sessions, upload question papers and their answers, produce OMR sheets, assign evaluation duties to staff members, and publish the results. The staff members assist with exam administration tasks such as uploading question papers, creating OMRs, and managing students and applications. Once authorized by the administrator, the assessors can use the Flutter application to scan and assess the OMR sheets that have been assigned to them.

ExamAIR's CNN-based OMR evaluation module, which is based on OpenCV's image processing methods, is one of its main features. This finds the filled bubbles in the scanned OMR sheets, uses intelligent selection logic to validate the answers, compares the answers to the uploaded answer key, calculates the scores, and stores the evaluation results in the OMR system database. The hybrid deep learning and image processing approach can improve evaluation accuracy and lower the likelihood of human error.

Additionally, the system provides automated result processing, centralized examination monitoring, role-based access control, secure authentication, and PDF upload and management. ExamAIR ultimately provides educational institutions and examination authorities with an effective, scalable, and dependable digital solution that guarantees speed, accuracy, and operational efficiency while streamlining and improving the management and evaluation of large-scale, OMR-based exams.

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1.INTRODUCTION

1.1 INTRODUCTION

In this period of shifting educational and examination conditions, large-scale and effective exams have proven to be a major problem for educational institutions and examination bodies. There is a lot of effort involved in managing question papers, delivering OMR sheets, assessing responses, and processing the findings in typical OMR examination systems. They are laborious, prone to human error, and difficult to do efficiently in a large-scale assessment. In light of these issues, **ExamAIR** was created as a comprehensive digital OMR-based examination management and evaluation system that completes the task while automating the entire procedure.

ExamAIR provides an effective and dependable examination evaluation solution thanks to its comprehensive platform, which includes the web, mobile apps, REST API, deep learning models, and image processing. Along with improved accuracy, speed, and centralized control over examination activities, the system has a strong emphasis on automating the creation of OMR sheets, managing answer keys, assigning evaluators, scanning OMR sheets, and publishing results.

The Flutter mobile app and Python Django backend used in the development of ExamAIR can accommodate many user roles, including Administrator, Employee, and Evaluator. Test creation, employee registration, test sessions, exam centers, question paper and answer key uploads, evaluator approval, OMR sheet assignment, and publication of exam results are all under the admin's purview. With the assistance of the staff, the administrator oversees the examination-related operations, and the evaluators use the Flutter app to scan the assigned OMR sheets after receiving the administrator's clearance.

ExamAIR's CNN-based OMR evaluation system and OpenCV image processing methods are key components. This system automatically scans OMR sheets from scanned documents, finds filled bubbles, compares the filled bubbles to the supplied answer key, calculates exam scores, and stores evaluation results in the database. The use of deep learning and image processing technologies significantly improves evaluation accuracy while lowering manual labor and processing time.

For easy and safe test handling, the platform also offers secure authentication, role-based access control, centralized data administration, REST API connectivity, PDF upload and management, and automatic result creation. ExamAIR offers a scalable and reliable solution for efficiently administering

and conducting digital exams by fusing the newest software with clever OMR evaluation methods. ExamAIR's architecture, modules, functions, technologies, and practical features are covered in this paper along with how it may enhance test administration, automate OMR review, and boost the effectiveness and dependability of exam processing systems.

1.2 EXISTING SYSTEM

In the current examination system, many institutions still rely on manual methods for conducting and evaluating OMR-based examinations. Processes such as OMR sheet distribution, evaluator assignment, answer checking, and result preparation are often handled manually, making the system slow and error-prone.

Most existing systems lack automation, centralized management, and intelligent OMR evaluation capabilities. Administrators usually use separate tools for managing examinations, storing answer keys, and publishing results, which leads to poor coordination and increased workload. Manual verification of OMR sheets also consumes significant time and may reduce evaluation accuracy.

Additionally, traditional systems generally do not support mobile-based OMR scanning, automated rank list generation, secure role-based access, or real-time result publication. These limitations make existing examination systems inefficient for handling large-scale examinations effectively.

1.3 PROPOSED SYSTEM

ExamAIR aims to break these limitations on the traditional examination systems by introducing a modern digital OMR based examination evaluation system to automate the management of examinations, the processing of OMR and the generation of examination results. It comprises a Django-based web app, Flutter mobile app, REST APIs, and a CNN-based OMR evaluation model, creating a unified and streamlined examination management system.

The system provides a secure web interface for the admins to manage examinations, sessions, centres, employees, question papers, answer keys, and evaluators assignments. Activities related to examination can be carried out with the help of employees, while evaluators can scan and evaluate OMR sheets with the help of the mobile application.

The automated OMR evaluation process for ExamAIR is a significant feature that is done by OpenCV image processing and CNN deep learning methods. The system automatically marks bubbles, checks against the answer book, scores and saves the results in the database. It also facilitates the automatic generation of the rank list and publishing of results.

The proposed system will enhance the accuracy of the evaluation, decrease the manual labor and decrease the time of processing results, and to ensure secure, centralized administration of examinations for large scale OMR-based examinations.

2. SYSTEM REQUIREMENTS AND SPECIFICATIONS

To develop an efficient and reliable digital OMR examination evaluation system, proper system requirement analysis and specification are essential. The requirement specification defines the functionalities and system behaviour needed to meet the operational requirements of examination authorities, administrators, employees, and evaluators. ExamAIR is designed to automate examination management, OMR processing, and result generation while ensuring accuracy, security, and ease of use.

2.1 FUNCTIONAL REQUIREMENTS

1. Examination Management Module

This module enables the admin to manage the complete examination process efficiently. The admin can create examinations, define examination sessions such as morning or afternoon sessions, and manage examination centres. The system allows uploading of question papers and answer key PDFs based on specific examination sessions. It also supports downloading question papers whenever required. This module helps organize examinations systematically and reduces manual management effort.

2. Employee Management Module

The Employee Management Module allows the admin to register and manage employees who assist in examination-related operations. Employees can log in to the system and access the examinations assigned to them. They can upload question papers and answer keys, generate OMR sheets, and view student and application details. This module improves coordination between the admin and supporting staff while ensuring controlled access to system functionalities.

3. OMR Sheet Generation Module

This module is responsible for generating OMR sheets for examinations automatically. The system creates standardized OMR sheets based on the examination details and session information. Generated OMR sheets can be downloaded and distributed to candidates for examination purposes. Automating the OMR generation process reduces errors and saves significant administrative time.

4. Evaluator Management Module

The Evaluator Management Module manages evaluator registration and approval processes through the Flutter mobile application. Evaluators can register using the application, after which a request is sent to the admin for approval. Only approved evaluators can access the system and perform evaluation activities. The admin can assign specific OMR sheet ranges or examinations to evaluators, ensuring organized and secure evaluation management

5. OMR Evaluation Module

The OMR Evaluation Module is one of the core functionalities of ExamAIR. Evaluators can scan assigned OMR sheets using the mobile application. The scanned images are processed using OpenCV image processing techniques and a CNN-based deep learning model to detect filled bubbles accurately. The system automatically identifies selected answers, compares them with the stored answer key, calculates scores, and generates evaluation results. This module significantly reduces manual evaluation work and improves result accuracy.

6. Result Management Module

This module handles the storage, review, and publication of examination results. After evaluation, the calculated results are stored securely in the database. The admin can review results and publish them through the system. The module enables fast and accurate result processing while maintaining centralized result management.

7. Student and Application Management Module

The system provides functionalities for viewing and managing student details and examination applications. Admins and employees can access candidate information, application status, and examination-related records. This module helps maintain organized candidate data and simplifies examination administration activities.

8. Authentication and Access Control Module

The Authentication and Access Control Module ensures secure system access for all users. The system supports role-based access control for admins, employees, and evaluators. Users can securely log in and access only the functionalities assigned to their role. Evaluator accounts require admin approval before activation, enhancing overall system security and controlled access management.

9. REST API Integration Module

The system uses REST APIs for communication between the Django backend, Flutter mobile application, and the CNN-based OMR evaluation model. APIs enable secure and efficient data transfer across different modules and support seamless integration between the web and mobile platforms.

10. Database Management Module

This module manages the storage and retrieval of examination data, user information, question papers, answer keys, OMR evaluation results, and other system records. The database ensures data consistency, reliability, and centralized management of all examination-related information.

2.2 NON-FUNCTIONAL REQUIREMENTS

1. Performance

The ExamAIR system should deliver efficient and responsive performance during examination management and OMR evaluation operations. Activities such as uploading question papers, generating OMR sheets, scanning answer sheets, calculating marks, and publishing results should be completed within a reasonable time to provide a smooth user experience, especially during large-scale examinations.

2. Security

ExamAIR must ensure secure handling of examination data, user information, answer keys, and evaluation results. The system uses secure authentication and role-based access control so that only authorized users can access specific functionalities. Sensitive information should be protected from unauthorized access and data loss.

3. Usability

ExamAIR should provide a simple and easy-to-use interface for administrators, employees, and evaluators. Users with basic computer knowledge should be able to manage examinations, upload files, scan OMR sheets, and access results without complexity.

4. Compatibility

The application should function correctly across different devices and modern web browsers. The Flutter-based mobile application should operate efficiently on Android devices used for OMR scanning and evaluation activities.

5. Reliability

The system should produce accurate OMR evaluations, score calculations, and result generation consistently. Examination records and evaluation data must be stored securely to maintain data integrity and prevent information loss.

6. Maintainability

The software should follow a modular and organized architecture to simplify maintenance and future enhancements. Well-structured code and proper documentation should help developers update features, resolve issues, and improve system functionality efficiently.

7. Scalability

ExamAIR should support increasing numbers of users, examinations, OMR sheets, and evaluation records without significant reduction in performance. The system architecture must be capable of supporting future expansion and large-scale examination management requirements.

2.3 HARDWARE REQUIREMENTS

- Processor : Intel Core i3 or higher
- RAM : 8 GB minimum
- Storage : 8 GB free disk space or higher
- Internet Connection : Required
- Camera/Webcam : Mobile camera or webcam for OMR sheet scanning and evaluation

2.4 SOFTWARE REQUIREMENTS

- Operating System : Windows 7 or higher / macOS / Linux / Android
- Programming Language : Python, Dart, JavaScript, HTML, CSS

- Backend Framework : Django, Django REST Framework
- Frontend Framework : Flutter
- Machine Learning Libraries : TensorFlow / PyTorch, OpenCV, NumPy
- Database : MySQL
- Tools Used : Visual Studio Code, Android Studio, Postman, GitHub
- Version Control : Git and GitHub
- Browser : Google Chrome / Firefox / Microsoft Edge (latest versions)
- API Technology : REST API

SOFTWARE DESCRIPTION

A) Visual Studio Code

Visual Studio Code is a source code editor developed by Microsoft that supports multiple programming languages and development tools. It provides features such as debugging, syntax highlighting, code completion, and version control integration. In the ExamAIR project, Visual Studio Code is used for developing backend services, frontend components, and Flutter application modules efficiently.

B) Python

Python is a high-level programming language widely used for web development, automation, machine learning, and artificial intelligence applications. In ExamAIR, Python is used for backend programming and implementing the CNN-based OMR evaluation module. Its extensive libraries simplify image processing, API handling, and machine learning operations.

C) Django

Django is a Python-based web framework used for building secure and scalable web applications. It follows the Model-View-Template (MVT) architecture and includes features such as authentication, database management, and admin support. In ExamAIR, Django is used to develop the backend system, manage examination data, implement REST APIs, and handle user authentication and result storage.

D) Flutter

Flutter is an open-source UI toolkit developed by Google for building cross-platform mobile applications using a single codebase. In the ExamAIR project, Flutter is used to develop the evaluator mobile application, enabling evaluators to log in, scan OMR sheets, and access evaluation results through an easy-to-use interface.

E) OpenCV

OpenCV is an open-source computer vision and image processing library used for analyzing and processing digital images. In ExamAIR, OpenCV is used for preprocessing scanned OMR sheets, detecting answer bubbles, cropping answer regions, and improving image quality before passing the data to the CNN model for evaluation.

F) CNN (Convolutional Neural Network)

CNN is a deep learning technique mainly used for image recognition and classification tasks. In ExamAIR, the CNN model is used to classify filled and unfilled bubbles from scanned OMR sheets. The model improves answer detection accuracy and supports automated OMR evaluation.

G) MySQL

MySQL is a relational database management system used for storing and managing structured data. In ExamAIR, MySQL stores user details, examination information, question papers, answer keys, OMR evaluation results, and rank list data securely and efficiently.

H) REST API

REST API is used for communication between different modules of the system. In ExamAIR, REST APIs enable data exchange between the Django backend, Flutter mobile application, and OMR evaluation module, ensuring smooth integration of system components.

I) Postman

Postman is a software tool used for API development and testing. It allows developers to send requests, analyze responses, and validate API functionality. In ExamAIR, Postman is used to test backend APIs and verify proper communication between modules.

J) Git

Git is a distributed version control system used for tracking source code changes and managing project versions. In ExamAIR, Git is used for maintaining code history, handling updates, and supporting collaborative development.

K) GitHub

GitHub is a cloud-based platform used for hosting and managing Git repositories. It supports version control, collaboration, and code backup. In ExamAIR, GitHub is used to maintain the project repository and securely store source code throughout the development process.

3. SYSTEM DESIGN

3.1 MODULE DESCRIPTION

The ExamAIR system consists of multiple integrated modules designed to automate and simplify examination management, OMR evaluation, and result processing. These modules work together to provide efficient examination handling for administrators, employees, and evaluators. The platform offers centralized management, secure access, automated OMR processing, and accurate result generation. The core modules include the Examination Management Module, Employee Management Module, OMR Evaluation Module, Result Management Module, and Admin Control Module.

3.1.1 Examination Management Module

Module functions are:

- Allows admins and employees to create and manage examinations efficiently.
- Supports creation of examination sessions such as morning and afternoon sessions.
- Enables management of examination centres and examination details.
- Provides functionality to upload question paper PDFs and answer key PDFs based on examination sessions.
- Supports automatic OMR sheet generation for examinations.
- Allows downloading of question papers and generated OMR sheets.
- Maintains organized examination records for easy access and monitoring.

3.1.2 Employee Management Module

Module functions are:

- Allows the admin to register and manage employees who assist in examination operations.
- Employees can log in securely and access assigned examinations.
- Supports uploading of question papers and answer key PDFs by employees.
- Enables employees to generate OMR sheets and manage examination-related activities.
- Allows viewing of student details and examination applications.
- Ensures role-based access control for secure employee operations.

3.1.3 OMR Evaluation Module

Module functions are:

- Allows evaluators to register through the Flutter mobile application.
- Sends evaluator registration requests to the admin for approval.
- Provides secure evaluator login after admin approval.
- Allows evaluators to view assigned OMR sheet ranges and examination details.
- Supports scanning of OMR sheets using the mobile device camera or webcam.
- Uses OpenCV image processing techniques for image preprocessing and bubble detection.
- Uses a CNN-based deep learning model to identify filled bubbles accurately.
- Automatically compares detected answers with the uploaded answer key.
- Calculates examination scores and stores evaluation results in the database.
- Allows evaluators to review scanned results before submission.

3.1.4 Result and Rank List Management Module

Module functions are:

- Automatically generates examination results after OMR evaluation.
- Calculates candidate scores accurately based on the answer key.
- Generates rank lists according to candidate performance.
- Allows admins to review and publish examination results and rank lists.
- Maintains centralized result records for future reference.
- Supports fast and transparent result publication.

3.1.5 Admin Panel Module

Module functions are:

- Provides centralized control and monitoring features for administrators.
- Includes a dashboard displaying important system information such as:
 - Total examinations
 - Total employees
 - Total evaluators
 - Total students
 - Evaluation progress

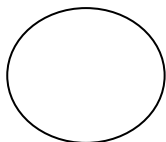
- Published results
- Recent activities
- Allows admins to:
 - Manage employees
 - Approve evaluator requests
 - Manage examinations and sessions
 - Assign OMR sheet ranges to evaluators
 - Monitor evaluation activities
- Maintains activity logs for tracking important system operations.
- Supports secure admin login, logout, and password management.
- Ensures role-based access control for secure administration and data management.

3.2 DATA FLOW DIAGRAM

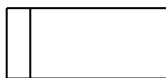
A data flow diagram is a design tool used for developing flowchart. It indicates the data flow through the system. This is constructed by using the symbols listed below:



Represents source/destination data



Represents process



Represents data store



Represents dataflow

By utilizing a Data Flow Diagram (DFD), developers can understand the functioning of a system, its intended outcomes, and the implementation methods. The location and method of data processing within the system are also critical.

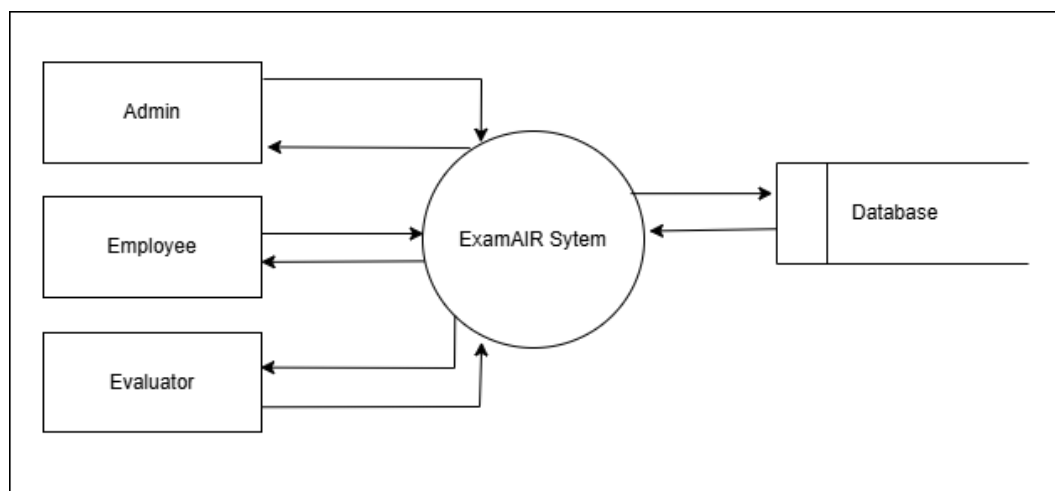
DFD - LEVEL 0

Fig 1 – DFD Level 0

The Level-0 Data Flow Diagram (DFD) for the ExamAIR – Digital OMR Examination Evaluation System is shown above. This diagram presents a high-level overview of how external entities interact with the system and how the system communicates with the central database.

Admins interact with the ExamAIR system through the web application to manage the complete examination process. They can create exams, sessions, and centres, register employees, approve evaluator registration requests, upload question papers and answer keys, generate OMR sheets, assign OMR sheets to evaluators, and publish examination results. The admin also monitors and controls overall system activities.

Employees support the examination management process through the web portal. They can add questions and answer keys, generate question papers, create OMR sheets, and assist in managing examination-related operations and data entry.

Evaluators interact with the system using the mobile application. They can register through the app and access the system only after admin approval. Evaluators can view assigned OMR sheets, scan candidate OMR answer sheets using the mobile camera, review detected answers, verify evaluation results, and submit the evaluated data to the system.

The ExamAIR system acts as the central processing unit that coordinates all examination activities, including exam management, OMR generation, evaluator assignment, OMR scanning, answer evaluation, and result publishing. The system ensures smooth communication between the web application, mobile application, and the database.

The Database stores all system-related information, including user details, employee and evaluator records, exams, sessions, centres, question papers, answer keys, OMR sheet data, scanned responses, evaluator assignments, calculated scores, and published results. This centralized storage ensures secure data management and efficient retrieval of examination information.

DFD – LEVEL 1

ExamAIR System

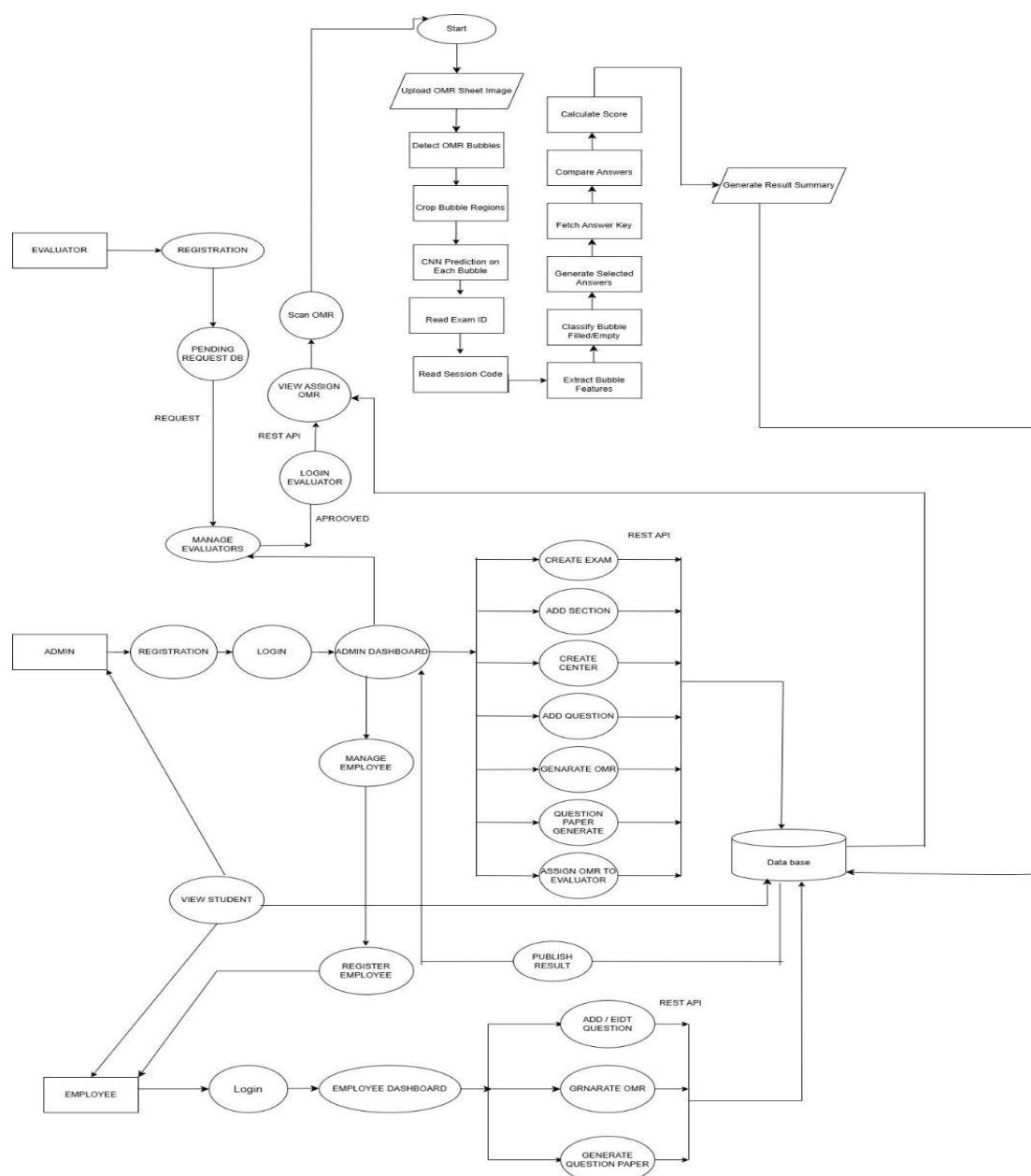


Fig 2 – DFD Level1

DFD – LEVEL 2

OMR Evaluation Module

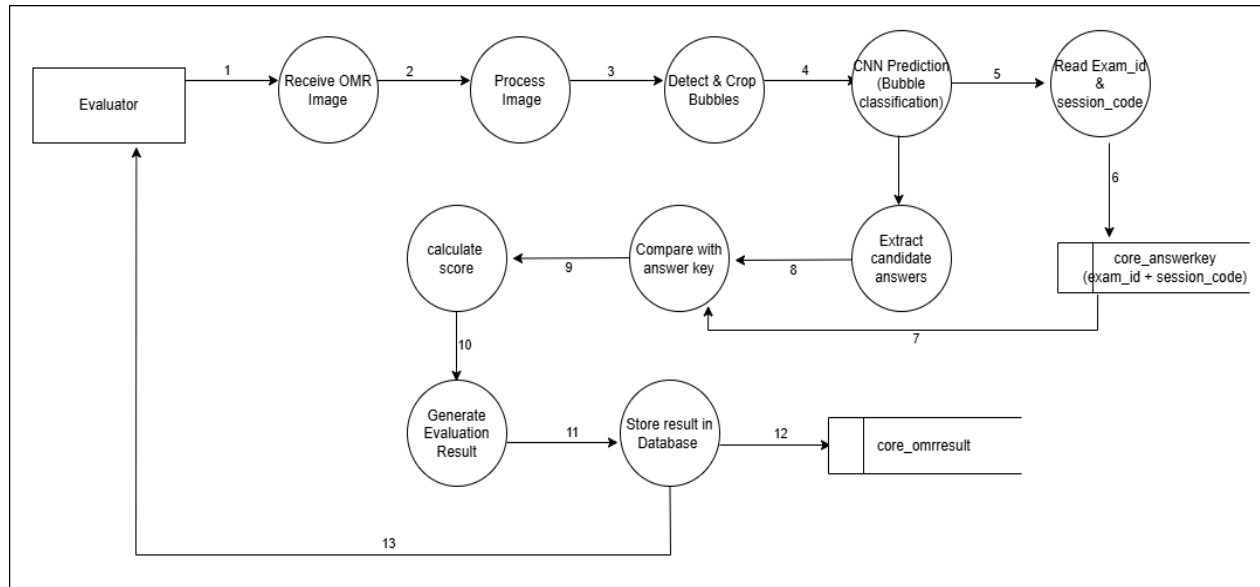


Fig 3 – DFD Level 2 (OMR Evaluation Module)

- 1 – Upload OMR sheet
- 2 – OMR Image
- 3 – processed Image
- 4 – Bubble images
- 5 – Predicted values
- 6 – exam id, session code
- 7 – Answer key
- 8 – Candidate answers
- 9 – Comparison Data
- 10 – Score & Statistics
- 11 – Result Data
- 12 – Save Result
- 13 – Review evaluation result

OMR Assignment Module

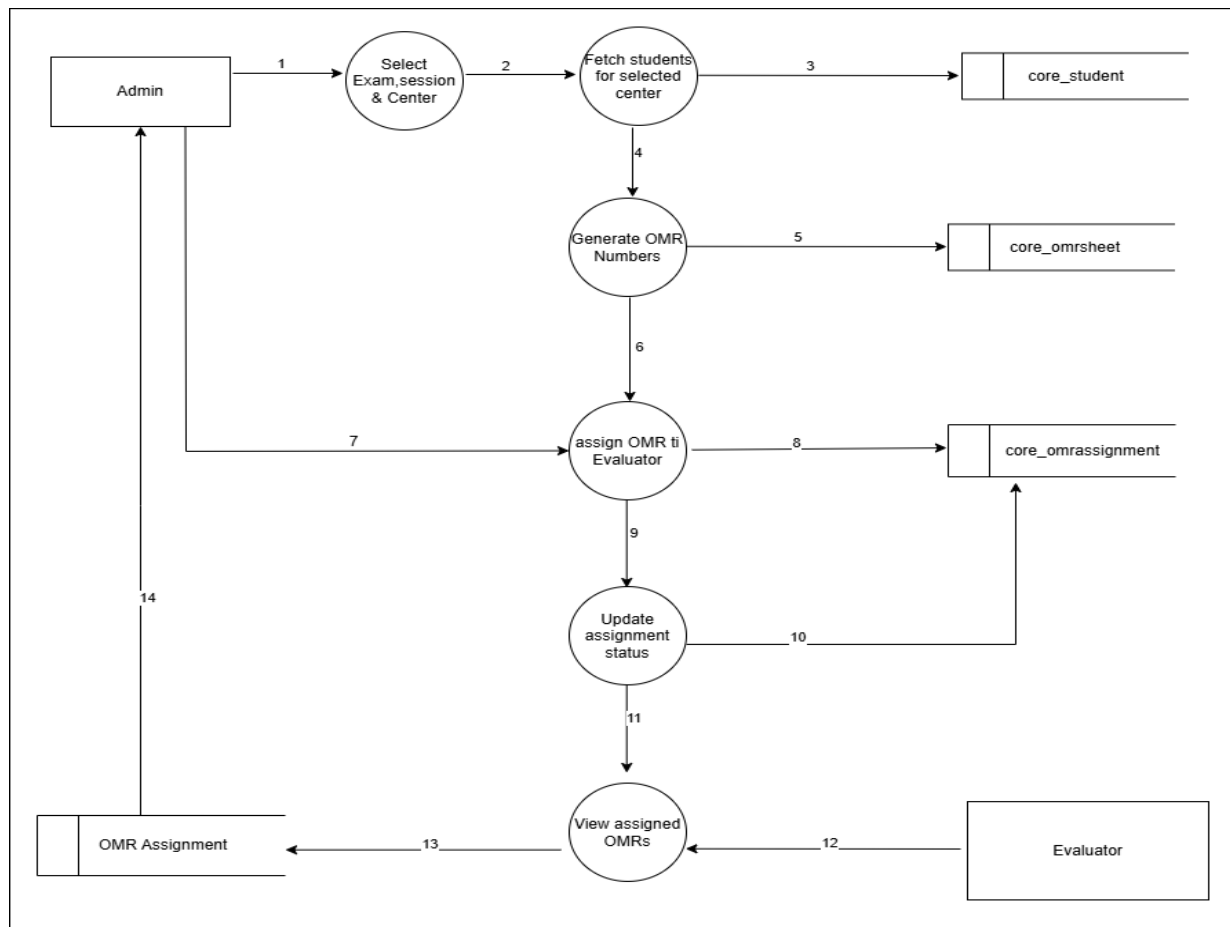


Fig 4 – DFD Level 2 (OMR assignment module)

- 1 – Select Exam, Session, Center
- 2 – Selection Data
- 3 – Student List
- 4 – Students
- 5 – OMR Numbers
- 6 – Generated OMR Numbers
- 7 – Assign Evaluator
- 8 – Assignment Data
- 9 – Assignment confirmation

- 10 – Update status
- 11 – Updated assignment
- 12 – Fetch assigned OMR
- 13 – Assigned OMR list
- 14 – Assignment summary

3.3 USE CASE DIAGRAM

Admin

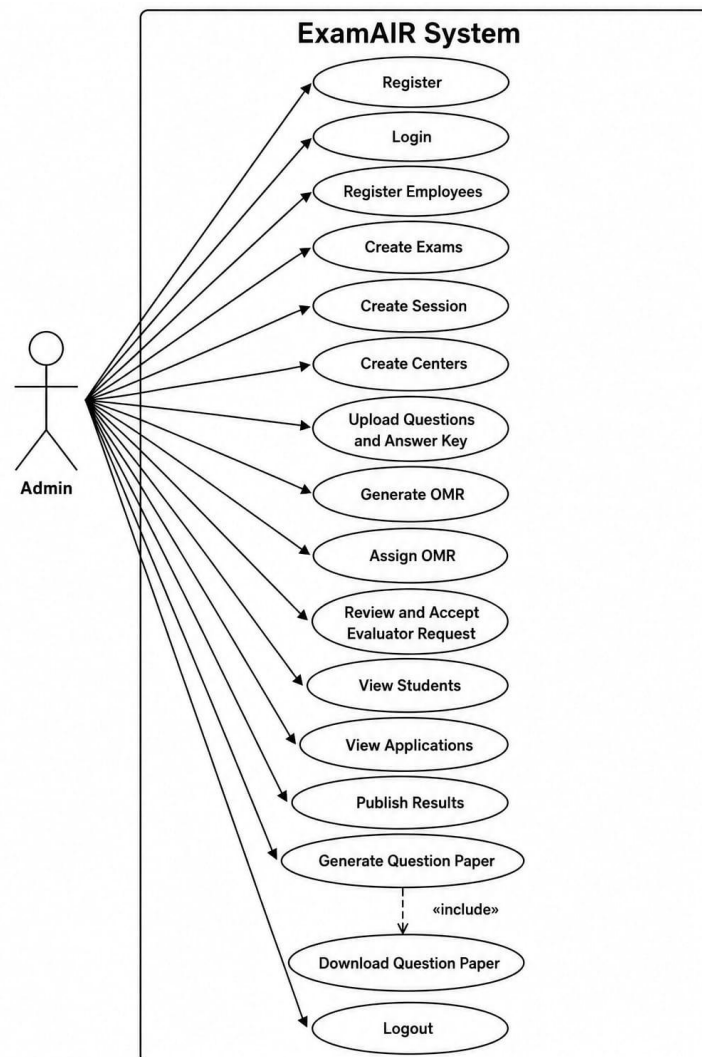


Fig 5 – use case diagram (admin)

Employee

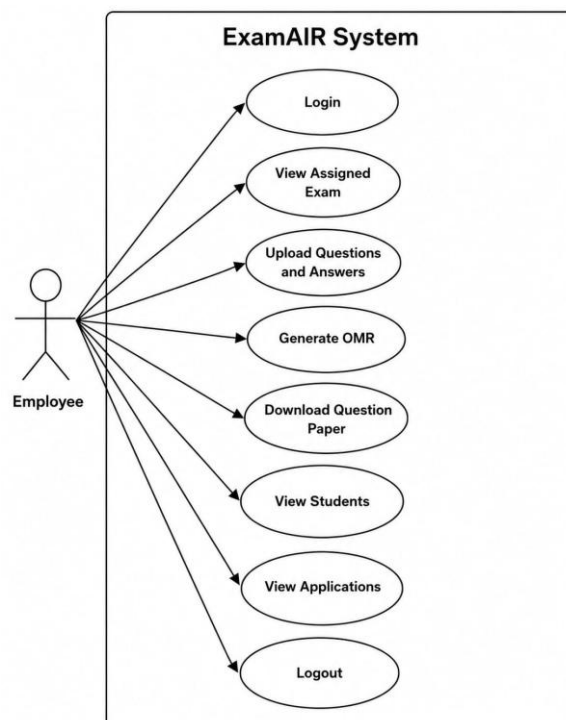


Fig 6 – use case diagram (employee)

Evaluator

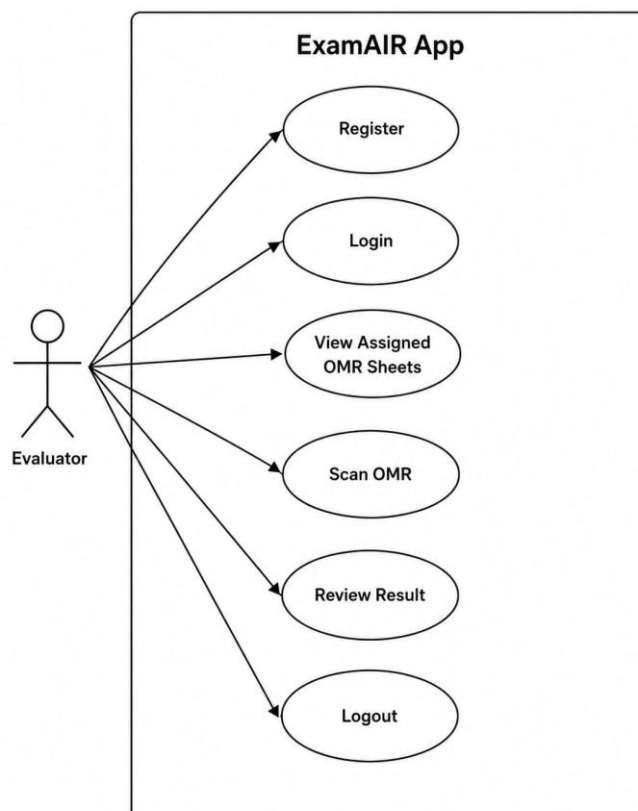


Fig 7 – use case diagram (evaluator)

3.4 E-R DIAGRAM

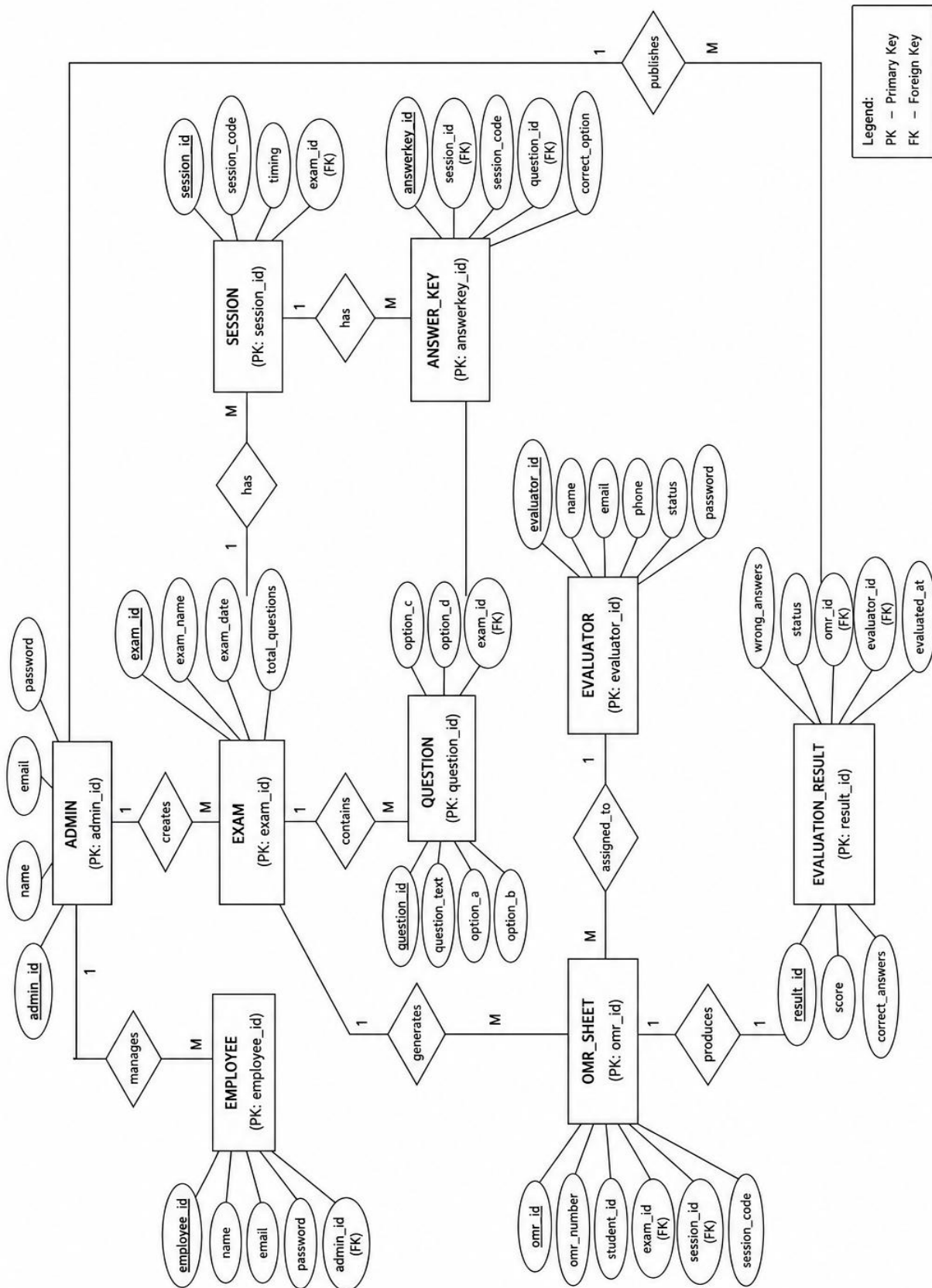


Fig 8 – ER diagram

1 : 1 – ONE TO ONE

1 : M – ONE TO MANY

M : N – MANY TO MANY

3.5 TABLE DESIGN

1. Table Name: core_user

Description : Used to store login and account details of admins, employees, evaluators, and organization administrators.

Primary Key: id

SI No	Name	Type	Attribute
1	id	bigint	Primary
2	organization_id	bigint	Not Null
3	name	varchar(255)	Not Null
4	email	varchar(255)	Not Null
5	password	varchar(255)	Not Null
6	user_type	varchar(50)	Not Null
7	created_by_id	bigint	Null
8	status	varchar(20)	Not Null
9	created_at	datetime	Null

Table 1 – core_user

2. Table Name: core_organisation

Description : Used to store organisation or institution details conducting examinations.

Primary Key: id

SI No	Name	Type	Attribute
1	id	bigint	Primary

2	organisation_name	varchar	Not Null
3	address	text	Null
4	contact_number	varchar	Null
5	email	varchar	Null
6	created_at	datetime	Null

Table 2 – core_organisation

3. Table Name: core_student

Description : Used to store student details.

Primary Key: id

SI No	Name	Type	Attribute
1	id	bigint	Primary
2	student_name	varchar	Not Null
3	application_number	varchar	Not Null
4	email	varchar	Null
5	phone	varchar	Null
6	exam_id	bigint	Not Null
7	created_at	datetime	Null

Table 3 – core_student

4. Table Name: core_exam

Description : Used to store examination details.

Primary Key: id

SI No	Name	Type	Attribute
1	id	bigint	Primary
2	exam_name	varchar	Not Null

3	exam_code	varchar	Not Null
4	exam_date	date	Not Null
5	total_questions	int	Not Null
6	organisation_id	bigint	Not Null
7	created_at	datetime	Null

Table 4 – core_exam

5. Table Name: core_question

Description : Used to store examination question details.

Primary Key: id

SI No	Name	Type	Attribute
1	id	bigint	Primary
2	exam_id	bigint	Not Null
3	session_id	bigint	Not Null
4	question_number	int	Not Null
5	correct_option	varchar	Not Null
6	created_at	datetime	Null

Table 5 – core_question

6. Table Name: core_answerkey

Description : Used to store uploaded answer key details for examinations.

Primary Key : id

SI No	Name	Type	Attribute
1	id	bigint	Primary

2	exam_id	bigint	Not Null
3	session_id	bigint	Not Null
4	session_code	varchar(50)	Not Null
5	version	varchar(20)	Not Null
6	status	varchar(20)	Not Null
7	finalized	boolean	Not Null
8	published_at	datetime	Null
9	created_at	datetime	Null
10	updated_at	datetime	Null
11	answer_key_pdf	varchar(255)	Null
12	question_paper_pdf	varchar(255)	Null
13	answer_key_json	json	Null

Table 6 – core_answerkey

7. Table Name: core_examsession

Description : Used to store examination session details such as session timing, reporting time, and candidate allocation information.

Primary Key: id

Sl No	Name	Type	Attribute
1	id	bigint	Primary
2	exam_id	bigint	Not Null
3	session_name	varchar(100)	Not Null
4	session_code	varchar(10)	Not Null
5	session_date	date	Not Null
6	start_time	time	Not Null
7	end_time	time	Not Null
8	reporting_time	time	Null

9	gate_close_time	time	Null
10	max_candidates	int	Not Null
11	allocated_candidates	int	Default 0
12	difficulty_level	varchar(50)	Null

Table 7 – core_examsession

8. Table Name: core_omrsheet

Description : Used to store generated OMR sheet details.

Primary Key: id

SI No	Name	Type	Attribute
1	id	bigint	Primary
2	exam_id	bigint	Not Null
3	student_id	bigint	Not Null
4	omr_number	varchar	Not Null
5	omr_sheet_file	varchar	Null
6	generated_at	datetime	Null

Table 8 – core_omrsheet

9. Table Name: core_omrassignment

Description : Used to assign OMR sheet ranges to evaluators.

Primary Key: id

SI No	Name	Type	Attribute
1	id	bigint	Primary
2	evaluator_id	bigint	Not Null
3	exam_id	bigint	Not Null
4	start_range	int	Not Null

5	end_range	int	Not Null
6	assigned_at	datetime	Null

Table 9 – core_omrassignment

10. Table Name: core_omrresult

Description : Used to store OMR evaluation results generated by the CNN model.

Primary Key: id

Sl No	Name	Type	Attribute
1	id	bigint	Primary
2	student_id	bigint	Not Null
3	exam_id	bigint	Not Null
4	score	decimal	Not Null
5	evaluated_by	bigint	Null
6	result_status	varchar	Not Null
7	evaluated_at	datetime	Null

Table 10 – core_omrresult

11. Table Name: core_ranklist

Description : Used to store published rank list details.

Primary Key: id

Sl No	Name	Type	Attribute
1	id	bigint	Primary
2	student_id	bigint	Not Null
3	exam_id	bigint	Not Null
4	rank	int	Not Null
5	total_score	decimal	Not Null
6	published_at	datetime	Null

Table 11 – core_ranklist

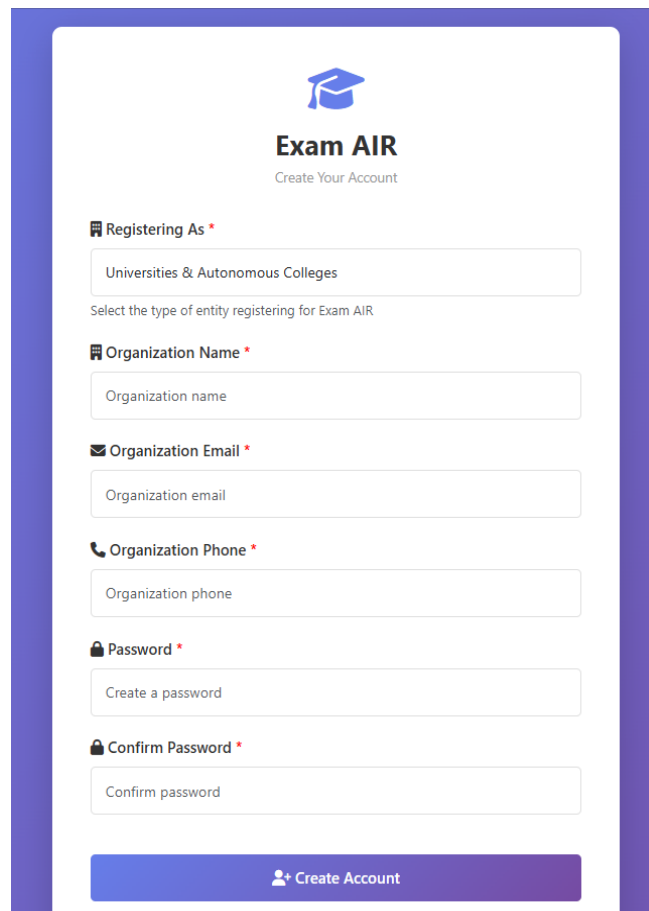
4. IMPLEMENTATION

4.1 Admin and Employee Login Page

Fig 9 - Admin and Employee Login Page

Figure 9 shows the login page of the ExamAIR web application. Administrators and employees can securely log in using their email address and password to access the examination management system. The module provides role-based access and secure authentication for authorized users.

4.2 Admin Registration Page



The screenshot displays the 'Exam AIR' admin registration interface. At the top, there is a blue graduation cap icon followed by the text 'Exam AIR' and 'Create Your Account'. Below this, the form is titled 'Registering As *' with a dropdown menu currently showing 'Universities & Autonomous Colleges'. A small instruction reads 'Select the type of entity registering for Exam AIR'. The subsequent fields are: 'Organization Name *' with a placeholder 'Organization name'; 'Organization Email *' with a placeholder 'Organization email'; 'Organization Phone *' with a placeholder 'Organization phone'; 'Password *' with a placeholder 'Create a password'; and 'Confirm Password *' with a placeholder 'Confirm password'. At the bottom of the form is a large blue button with a white user icon and the text 'Create Account'.

Fig 10 - Admin Registration Page

Figure 10 shows the admin registration page of the ExamAIR web application. Organizations and institutions can create an account by entering organization details such as name, email, phone number, and password. This module enables secure registration and account creation for authorized examination administrators.

4.3 Admin Dashboard

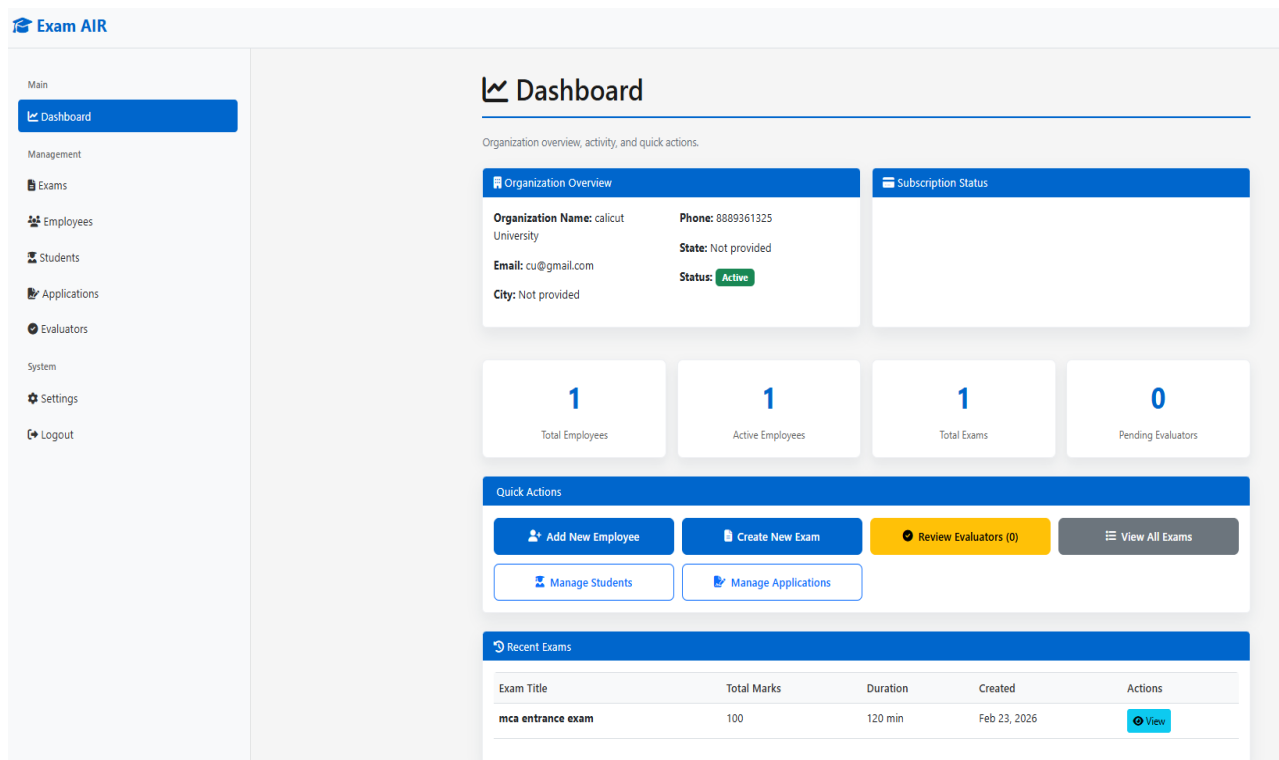


Fig 11 - Admin Dashboard

Figure 11 shows the admin dashboard of the ExamAIR system. The dashboard provides centralized access to examination management features such as employee management, exam creation, student management, evaluator review, and application management. It also displays organization details, examination statistics, and quick action options for efficient system administration.

4.4 Create Examination Module

The screenshot displays the 'Create New Exam' form in the ExamAIR system. The sidebar on the left includes a 'Main' section with a 'Dashboard' link, and a 'Management' section with links for 'Exams' (highlighted), 'Employees', 'Students', 'Applications', and 'Evaluators'. Below these is a 'System' section with 'Settings' and 'Logout' links. The main content area is titled 'Create New Exam' and features a form with the following fields: 'Title' (with placeholder 'Exam title'), 'Exam short name' (with placeholder 'Exam short name'), 'Exam code' (with placeholder 'Exam code (leave blank to auto-generate)'), 'Exam type' (a dropdown menu), 'Academic year' (with placeholder 'Academic year (e.g., 2025-2026)'), 'Level' (a dropdown menu), and 'Subjects' (currently showing 'null').

Fig 12 – Create examination module

Figure 12 shows the examination creation module of the ExamAIR system. Administrators can create new examinations by entering details such as examination title, exam code, exam type, academic year, level, and subjects. This module helps manage and organize examination information efficiently within the system.

4.5 Upload Question Paper and Answer Key Module

The screenshot displays the 'Upload Question Paper and Answer Key' form for the 'MSc CS entrance exam'. The sidebar on the left is identical to Figure 12, with 'Exams' selected. The main content area is titled 'Upload Question Paper and Answer Key: MSc CS entrance exam'. The form is divided into several sections: 'Session and Code' with a 'Session' dropdown (placeholder 'Select session') and a 'Code' input field (value 'A'); 'Question Paper PDF' with a 'Choose File' button and 'No file chosen' text; 'Answer Key PDF' with a 'Choose File' button and 'No file chosen' text; and 'Version and Status' with a 'Version' input field (value '1') and a 'Status' dropdown menu (value 'Provisional'). At the bottom right, there are 'Cancel' and 'Upload PDFs' buttons.

Fig 13 - upload Question Paper and Answer Key Module

Figure 13 shows the module used for uploading question papers and answer keys in the ExamAIR system. Administrators and employees can upload PDF files based on examination sessions and session codes. This module helps manage question papers and answer keys securely for automated OMR evaluation and result generation.

4.6 Session Creation Module

Sessions for MSc CS entrance exam

+ Create Session

Session Name: Session Label: Session date:

Start time: End time: Max candidates:

Reporting time: Gate close time: Status:

Difficulty level: Normalization factor: Paper Code (A-D):

Remarks:

Save Session

Existing Sessions

Name	Code	Paper Code	Date	Time	Max	Allocated	Status	Actions
morning	A	-	Feb. 11, 2026	7 a.m. - 8:40 a.m.	25	0	scheduled	Edit Upload PDFs
morning	B	-	Feb. 11, 2026	9 a.m. - 10:30 a.m.	25	0	scheduled	Edit Upload PDFs

Fig 14 - Session Creation Module

Figure 14 shows the session creation module of the ExamAIR system. Administrators can create examination sessions by entering session details such as session name, date, start time, end time, reporting time, maximum candidates, and paper code. This module helps organize examinations efficiently by managing multiple sessions and schedules for a single examination.

4.7 Centre Creation Module

Centres for MSc CS entrance exam ← Back

+ Create Centre

Centre code: Centre name: Centre type:

City: State: Total capacity:

Rooms: Seats per room: Status:

☐ Cctv available: ☐ Biometric available:

Save Centre

Existing Centres

Code	Name	City	State	Capacity	Status
ctr01	emscollege	kannur	kerala	25	active
ctr01	gck	kannur	kerala	25	active

Fig 15 - Centre Creation Module

Figure 15 shows the centre creation module of the ExamAIR system. Administrators can create and manage examination centres by entering details such as centre code, centre name, city, state, room availability, seating capacity, and centre status. This module helps organize examination centres and manage candidate allocation efficiently.

4.8 OMR Sheet Generation Module

Exam AIR cu@gmail.com

Generate OMR Sheets ← Back

Exam Details

Exam: MSc CS entrance exam Total Questions: 100 Existing OMR: 11

OMR Range

Session: Start Serial: End Serial:

Generate OMR Sheets

OMR sheets are generated for applications in the selected session that do not already have OMRs.

Fig 16 - OMR Sheet Generation Module

Figure 16 shows the OMR sheet generation module of the ExamAIR system. Administrators can generate OMR sheets by selecting the examination session and specifying the serial number range. The system automatically generates OMR sheets for registered candidates, which are later used for automated OMR evaluation and result processing.

4.9 OMR Assignment Module

Fig 17 - OMR Assignment Module

Figure 17 shows the OMR assignment module of the ExamAIR system. Administrators can assign a range of OMR sheets to evaluators by selecting the evaluator, session, and serial number range. This module helps distribute evaluation tasks efficiently among evaluators for automated OMR processing and result generation.

4.10 Question Paper Generation Module

mca entrance exam
Duration: 120 minutes Total Marks: 100 Total Questions: 200

Q1. Which keyword is used to define a function in Python? (Marks: 1.00)
A. func
B. define
C. def
D. lambda

Q2. Which keyword is used to define a function in Python? (Marks: 1.00)
A. func
B. define
C. def
D. lambda

Q3. Which SQL command is used to retrieve data? (Marks: 1.00)
A. GET
B. SELECT
C. FETCH
D. RETRIEVE

Q4. Which SQL command is used to retrieve data? (Marks: 1.00)
A. GET
B. SELECT
C. FETCH
D. RETRIEVE

Q5. Which protocol is used to transfer web pages? (Marks: 1.00)
A. FTP

Fig 18 - Question Paper Generation Module

Figure 18 shows the question paper generation module of the ExamAIR system. The system automatically generates and displays the examination question paper based on the uploaded questions and examination details. This module helps administrators and employees manage and print question papers efficiently for examination purposes.

4.11 Evaluator Approval Module

Evaluator Approval Workflow

Pending Approval (1)

Name	Email	Status	Registration Date	Actions
Evaluator One	evaluator1@example.com	Inactive	May 21, 2026 12:33	Review

Approved Evaluators (2)

Name	Email	Approved Date	Status
Evaluator Two	evaluator2@example.com	Jun 02, 2026	Active
Evaluator1	evaluator_01@example.com	Feb 23, 2026	Active

Fig 19 - Evaluator Approval Module

Figure 19 shows the evaluator approval module of the ExamAIR system. Administrators can review evaluator registration requests and approve or reject evaluators based on verification. Approved evaluators are activated and allowed to access the mobile application for OMR scanning and evaluation processes.

4.12 Student Management Module

Students

Student Records

Application No	Name	Phone	Email	Status	Registered
APP20260527114662	kavya bala	9448807713	-	registered	May 27, 2026
APP20260527783460	praveen	8906574321	-	registered	May 27, 2026
APP2026052747DD5F	ragu	1122334455	-	registered	May 27, 2026
APP20260527D8C738	ragu	1122334455	-	registered	May 27, 2026
APP2026030689E293	kavya	1122334455	-	registered	Mar 06, 2026
APP20260306453C52	kavya	1122334455	-	registered	Mar 06, 2026
APP202603067773E3	kavya	1122334455	-	registered	Mar 06, 2026
APP2026030680755D	kavya	1122334455	-	registered	Mar 06, 2026
APP202603056F016A	nandhana	8906574321	-	registered	Mar 05, 2026
APP202603056F6F76	nandhana	8906574321	-	registered	Mar 05, 2026
APP20260305DA971B	nandhana	8906574321	-	registered	Mar 05, 2026
APP202603050A28E0	nandhana	8906574321	-	registered	Mar 05, 2026
APP202603059F75B8	nandhana	8906574321	-	registered	Mar 05, 2026
APP202603051ED509	nandhana	8906574321	-	registered	Mar 05, 2026

Fig 20 - Student Management Module

Figure 20 shows the student management module of the ExamAIR system. Administrators and employees can view registered student details such as application number, student name, contact information, registration status, and registration date. This module helps manage and monitor student examination records efficiently.

4.13 Employee Creation Module

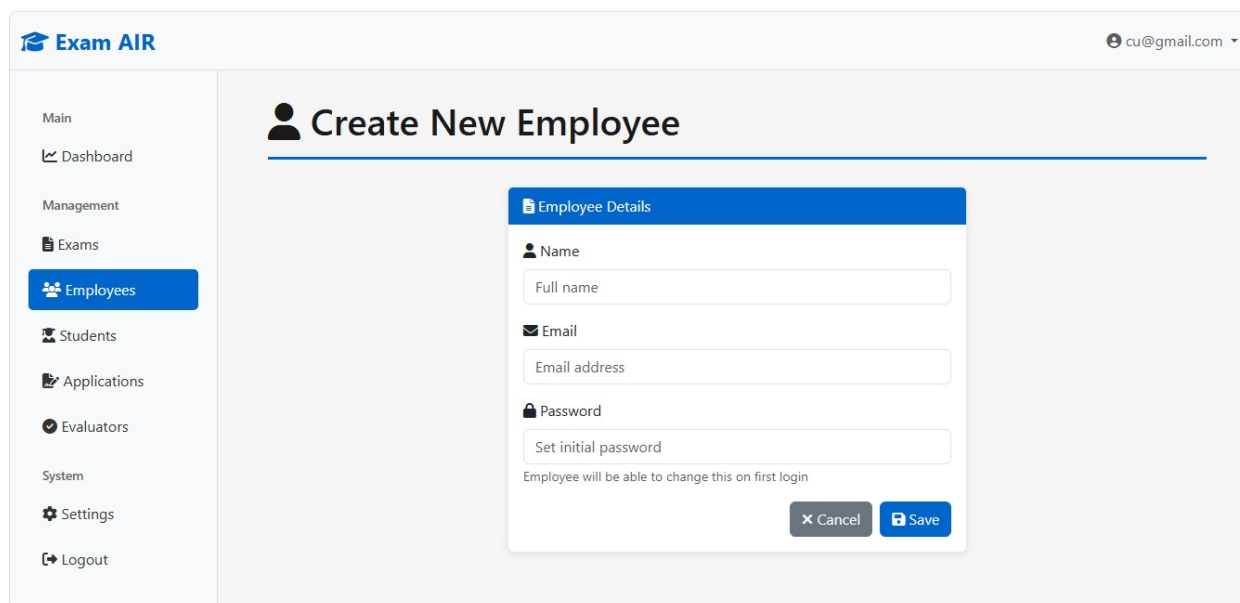


Fig 21 - Employee Creation Module

Figure 21 shows the employee creation module of the ExamAIR system. Administrators can create employee accounts by entering employee details such as name, email address, and password. This module helps manage authorized employees who assist in examination management and related operations within the system.

4.14 Employee Dashboard

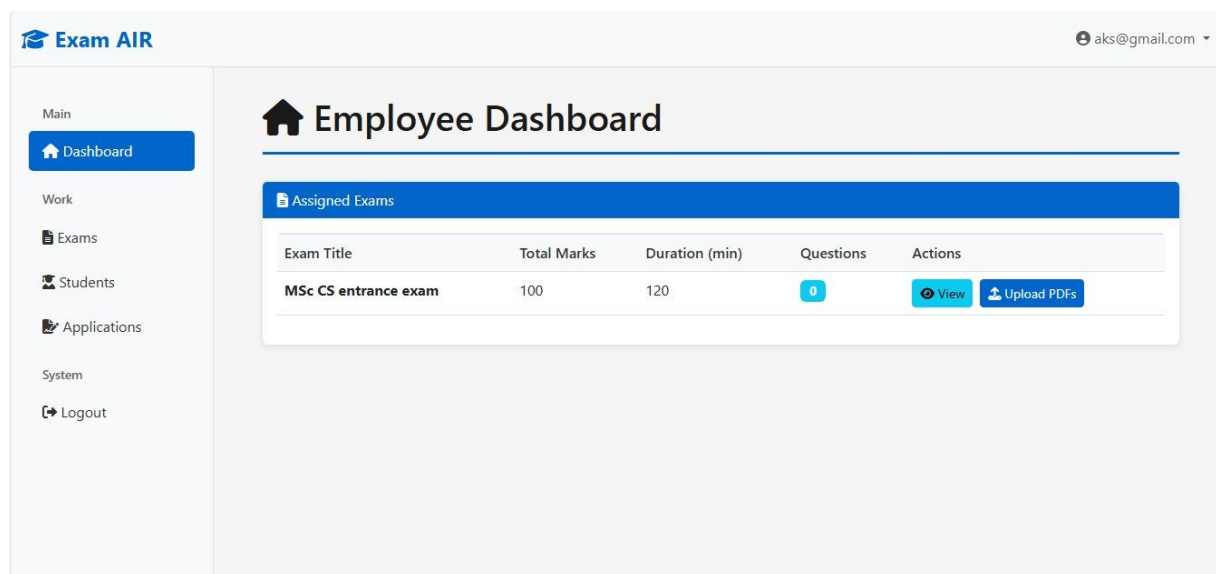


Fig 22 - Employee Dashboard

Figure 22 shows the employee dashboard of the ExamAIR system. Employees can view assigned examinations and perform examination-related tasks such as uploading question papers and answer keys, viewing student details, and managing examination records assigned by the administrator.

4.15 Examination Management Module

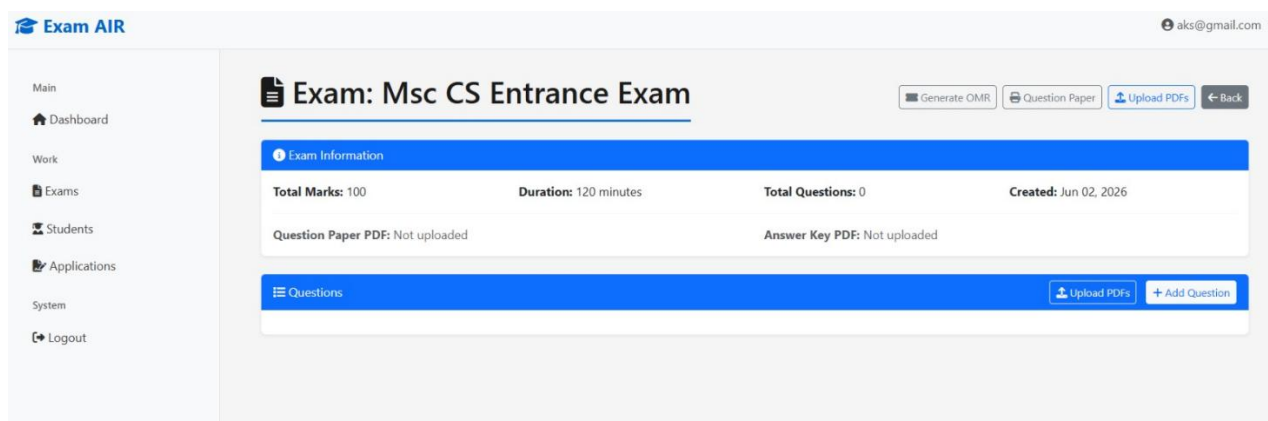
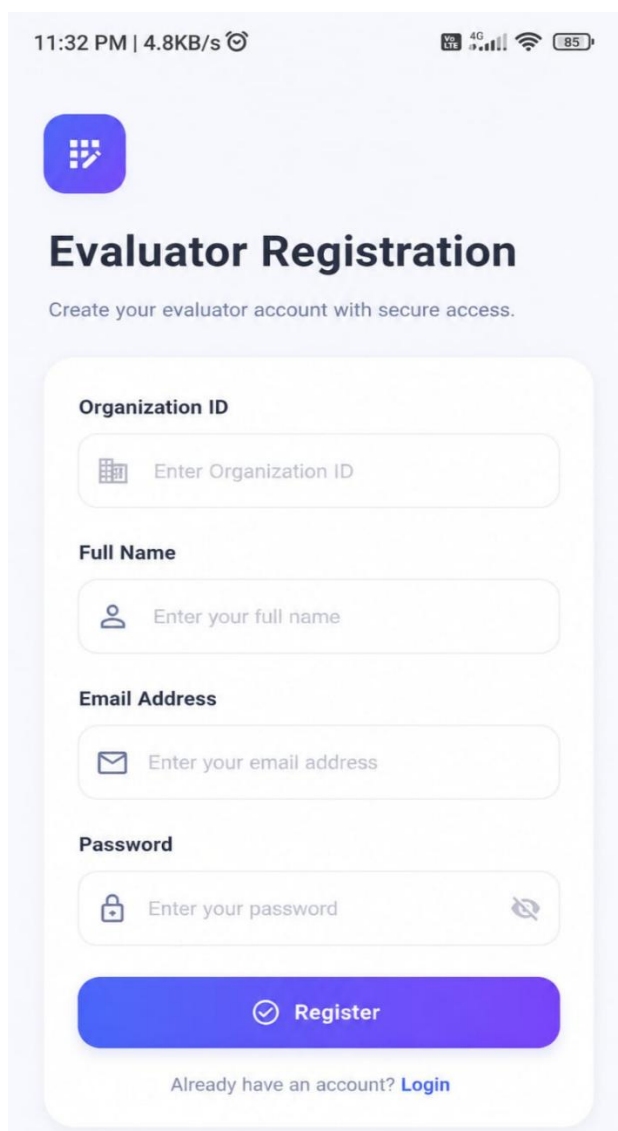


Fig 23 - Examination Management Module

Figure 23 shows the examination management module available for employees in the ExamAIR system. Employees can view assigned examination details, upload question papers and answer keys,

generate OMR sheets, and manage examination questions. This module helps employees perform examination-related operations efficiently within the system.

4.16 Evaluator Registration Module



The screenshot displays the 'Evaluator Registration' screen of the ExamAIR mobile application. At the top, the status bar shows the time as 11:32 PM, a data speed of 4.8KB/s, and icons for 4G, Wi-Fi, and a battery level of 85%. Below the status bar is a purple square icon with a white grid pattern. The title 'Evaluator Registration' is prominently displayed in bold black text, followed by the subtitle 'Create your evaluator account with secure access.' in a smaller, lighter font. The registration form consists of four input fields, each with a corresponding icon and placeholder text: 'Organization ID' with a building icon, 'Full Name' with a person icon, 'Email Address' with an envelope icon, and 'Password' with a lock icon. A blue 'Register' button with a white checkmark icon is positioned below the password field. At the bottom of the form, there is a link that says 'Already have an account? Login'.

Fig 24 - Evaluator Registration Module

Figure 24 shows the evaluator registration module of the ExamAIR mobile application. Evaluators can register by entering personal and organizational details such as name, email, and password. After registration, the request is sent to the administrator for approval before the evaluator can access the OMR evaluation system.

4.17 Evaluator Login Page

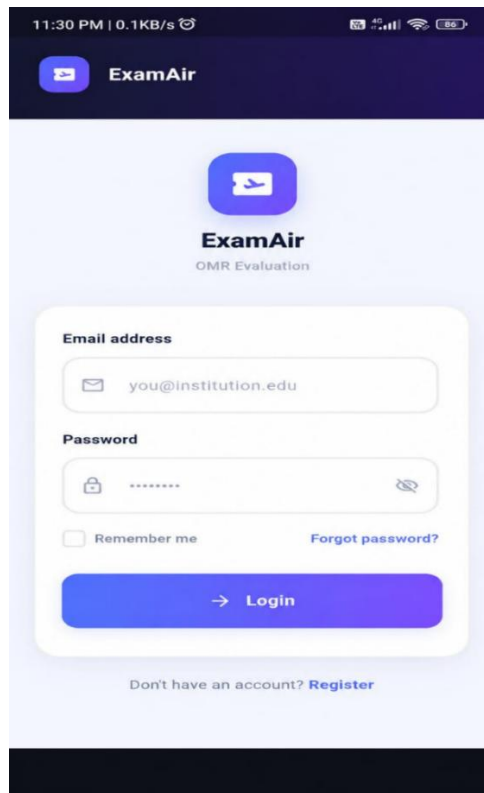


Fig 25 - Evaluator Login Page

Figure 25 shows the evaluator login page of the ExamAIR mobile application. Approved evaluators can securely log in using their registered email address and password to access the OMR evaluation system. This module provides secure authentication for evaluators before performing OMR scanning and evaluation tasks.

4.18 Evaluator Dashboard Module

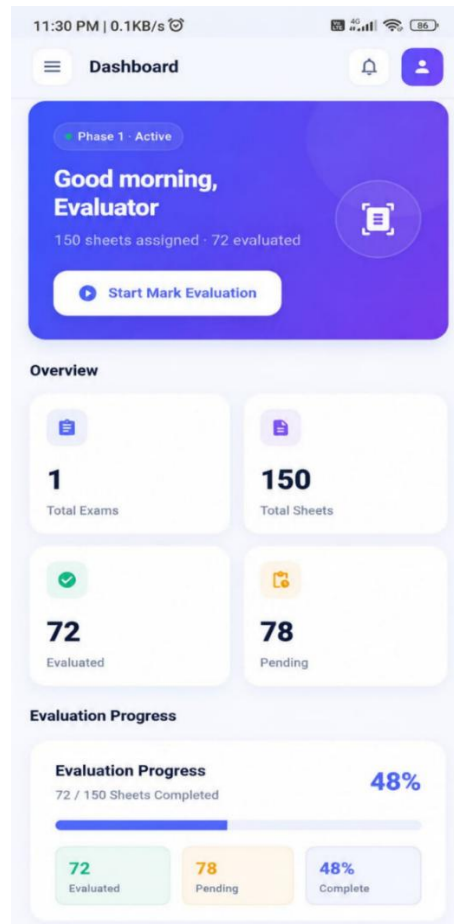


Fig 26 - Evaluator Dashboard Module

Figure 26 shows the evaluator dashboard of the ExamAIR mobile application. Evaluators can view assigned OMR sheets, completed evaluations, pending evaluations, and overall evaluation progress. This module helps evaluators efficiently manage and monitor the OMR evaluation process.

4.19 Assigned OMR Sheets Module

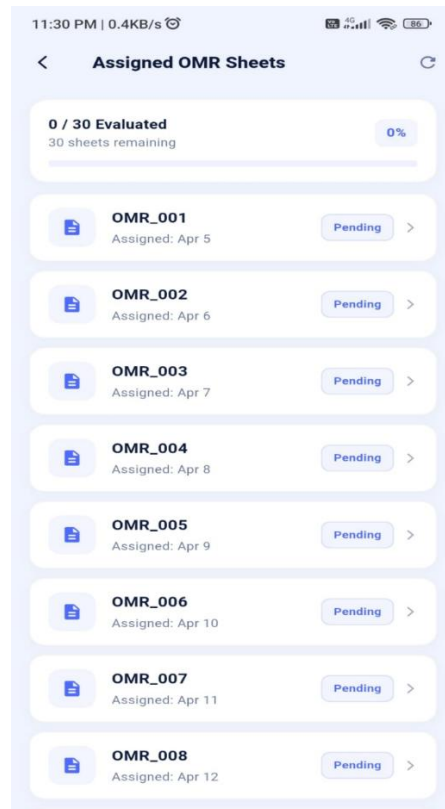


Fig 27 - Assigned OMR Sheets Module

Figure 27 shows the assigned OMR sheets module of the ExamAIR mobile application. Evaluators can view the list of assigned OMR sheets along with evaluation status and progress details. This module helps evaluators track pending and completed OMR evaluations efficiently.

4.20 OMR Scanning and Evaluation Module



Fig 28 - OMR Scanning and Evaluation Module

Figure 28 shows the OMR scanning and evaluation module of the ExamAIR mobile application. Evaluators can scan OMR sheets using the mobile camera by aligning the sheet within the scanning frame. The scanned OMR image is processed using OpenCV preprocessing techniques and evaluated using the CNN-based OMR evaluation model for automatic answer detection and score calculation.

4.21 OMR Evaluation Result Module

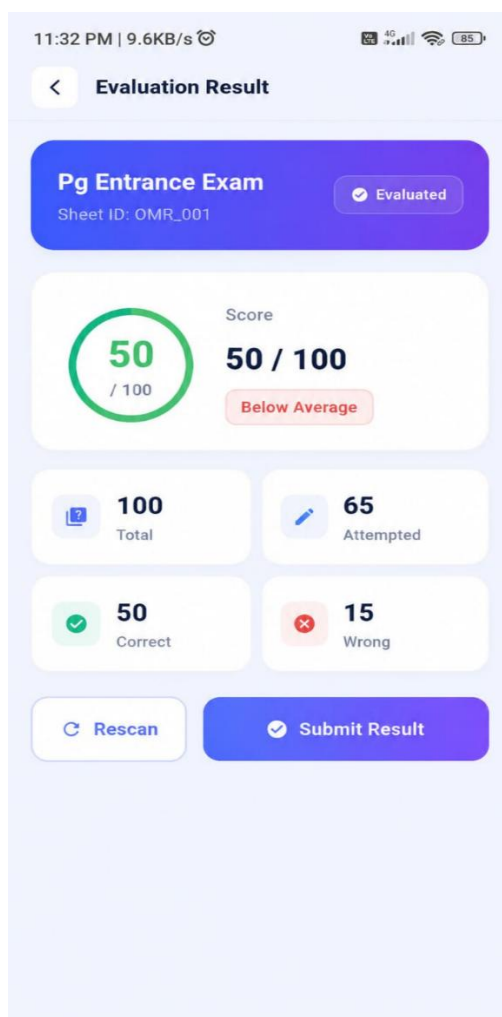


Fig 29 - OMR Evaluation Result Module

Figure 29 shows the OMR evaluation result module of the ExamAIR mobile application. After scanning and evaluating the OMR sheet, the system automatically displays the examination score, attempted questions, correct answers, wrong answers, and evaluation status. Evaluators can review the result and submit it to the system for final result processing and rank list generation

5. TESTING

Testing plays an essential role in the development of ExamAIR by ensuring that the system operates accurately, securely, and efficiently. Various testing techniques were carried out to detect issues, validate system functionality, and confirm that all modules work according to the project requirements. The testing phase helped improve the reliability and stability of the examination management and OMR evaluation system.

5.1 Unit Testing

Unit testing was performed to check the functionality of individual modules independently. Each function and component was tested separately to verify proper input processing, expected outputs, and error handling mechanisms.

The following modules were tested during unit testing:

- Admin and employee authentication
- Exam and session creation
- Question paper and answer key uploading
- OMR sheet generation
- CNN prediction functions
- Bubble detection and OMR scanning
- Score calculation and database updates
- Result generation and rank list preparation

This testing process helped identify programming errors at the initial stage and ensured that individual modules functioned correctly before integrating them into the complete system.

5.2 Integration Testing

Integration testing was conducted to verify proper communication between different modules of ExamAIR. The testing ensured smooth interaction between the frontend application, backend server, database, REST APIs, OpenCV image processing module, and CNN-based evaluation model.

The following operations were verified during integration testing:

- Data exchange between Flutter application and Django backend
- REST API request and response handling
- Database connectivity with examination modules
- OMR scanning and CNN evaluation workflow
- Answer key retrieval and score generation
- Result storage and rank list publishing
- Evaluator approval and assignment workflow

This testing confirmed that all integrated modules worked together properly without data transfer or communication problems.

5.3 User Acceptance Testing

User Acceptance Testing (UAT) was carried out to ensure that the system satisfies user requirements and operates effectively in real-time usage scenarios. The system was tested from the perspective of administrators, employees, and evaluators to verify usability and operational efficiency.

The following aspects were validated during UAT:

- Examination management operations
- User interface simplicity and usability
- Accuracy of OMR evaluation
- Result generation and rank list correctness
- Secure authentication and role-based access
- Mobile application functionality
- Overall system performance

Feedback obtained during the testing phase was used to improve usability and system functionality. Successful completion of User Acceptance Testing confirmed that ExamAIR is suitable for digital OMR-based examination management and automated evaluation processes.

5.4 Validation Testing

Validation testing was performed to verify whether the ExamAIR system satisfies all functional and non-functional requirements identified during the requirement analysis phase. This testing ensured that the developed system operates correctly, efficiently, and according to user expectations in real-time examination environments.

This testing ensures that:

- The system performs according to examination management requirements
- OMR evaluation results generated by the CNN model are accurate
- User authentication and role-based access control function properly
- Examination creation, session management, and OMR sheet generation work correctly
- Question paper and answer key upload functionalities operate properly
- OMR scanning and bubble detection processes function accurately
- Result generation and rank list publication are performed correctly
- REST API communication between frontend and backend works efficiently
- Flutter mobile application functions properly during evaluator operations
- Database storage and retrieval operations are accurate and reliable

Validation testing confirmed that the developed ExamAIR system is reliable, accurate, and suitable for real-world digital OMR-based examination management and automated evaluation processes.

5.5 TEST REPORT

TEST CASE 1: ADMIN TEST CASES

SL.NO	PROCESS	INPUT	EXPECTED OUTPUT	OBTAINED OUTPUT	REMARKS
1	Admin Login	Email, Password	Admin Homepage	Admin Homepage	Success
2	Admin Login	Invalid Email/Password	Error Message	Error Message	Success

3	Create Examination	Exam Details, Session Details	Examination Created Successfully	Examination Created Successfully	Success
4	Create Examination	Missing Required Fields	Error: "Fill all fields"	Error: "Fill all fields"	Success
5	Upload Question Paper	Valid PDF File	Question Paper Uploaded Successfully	Question Paper Uploaded Successfully	Success
6	Upload Answer Key	Valid PDF File	Answer Key Uploaded Successfully	Answer Key Uploaded Successfully	Success
7	Generate OMR Sheet	Valid Exam Details	OMR Sheet Generated Successfully	OMR Sheet Generated Successfully	Success
8	Approve Evaluator	Evaluator Request	Evaluator Approved Successfully	Evaluator Approved Successfully	Success
9	Publish Result	Evaluated Result Data	Results Published Successfully	Results Published Successfully	Success
10	Generate Rank List	Published Results	Rank List Generated Successfully	Rank List Generated Successfully	Success

Table 12 – admin test cases

TEST CASE 2: EMPLOYEE TEST CASES

SL.NO	PROCESS	INPUT	EXPECTED OUTPUT	OBTAINED OUTPUT	REMARKS
1	Employee Login	Email, Password	Employee Homepage	Employee Homepage	Success
2	Employee Login	Invalid Email/Password	Error Message	Error Message	Success
3	Upload Question Paper	Valid PDF File	Question Paper Uploaded Successfully	Question Paper Uploaded Successfully	Success
4	Upload Answer Key	Valid PDF File	Answer Key Uploaded Successfully	Answer Key Uploaded Successfully	Success
5	Generate OMR Sheet	Valid Exam Details	OMR Sheet Generated Successfully	OMR Sheet Generated Successfully	Success

6	View Student Details	Click View Students	Student Details Displayed	Student Details Displayed	Success
7	View Applications	Click View Applications	Application List Displayed	Application List Displayed	Success
8	Upload Invalid File	Unsupported File Format	Error Message	Error Message	Success
9	Missing Required Fields	Empty Input Fields	Error: "Fill all fields"	Error: "Fill all fields"	Success
10	Logout	Click Logout	Redirected to Login Page	Redirected to Login Page	Success

Table 13 – employee test cases

TEST CASE 3: EVALUATOR TEST CASES

SL.NO	PROCESS	INPUT	EXPECTED OUTPUT	OBTAINED OUTPUT	REMARKS
1	Evaluator Registration	Valid Registration Details	Request Sent to Admin	Request Sent to Admin	Success
2	Evaluator Login	Valid Credentials	Evaluator Homepage	Evaluator Homepage	Success
3	Evaluator Login	Login Before Approval	Approval Pending Message	Approval Pending Message	Success
4	View Assigned OMR	Click Assigned OMR	Assigned OMR List Displayed	Assigned OMR List Displayed	Success
5	Scan OMR Sheet	Valid OMR Image	Answers Detected Successfully	Answers Detected Successfully	Success
6	Scan OMR Sheet	Invalid/Blurred Image	Error Message	Error Message	Success
7	CNN OMR Evaluation	Filled Bubble Markings	Correct Answers Predicted	Correct Answers Predicted	Success
8	Generate Result	Evaluated OMR Data	Score Generated Successfully	Score Generated Successfully	Success
9	Save Result	Valid Result Data	Result Stored Successfully	Result Stored Successfully	Success
10	Logout	Click Logout	Redirected to Login Page	Redirected to Login Page	Success

Table 14 – evaluator test cases

5.6 MODEL PERFORMANCE EVALUATION

The OMR sheets were trimmed into OMR bubble images during dataset preparation so that the dataset could be trained. Three divisions of the dataset have been made:

Training, Validation, and Testing Datasets

The data distribution is displayed below.

Dataset for Training

Filled : 420 pictures

Empty : 420 pictures

Noise : 350 pictures

Verification Dataset

Filled : 90 pictures

Empty : 90 pictures

Noise : 75 pictures

Dataset for Testing

Filled : 90 pictures

Empty : 90 pictures

Noise : 75 pictures

In order to improve the model's performance when evaluated in real time, the data set included bubble samples taken in a variety of lighting circumstances, marking styles, and image qualities.

Before training, the data was preprocessed using a number of OpenCV techniques, including grayscale conversion, resizing, thresholding, contour extraction, and bubble segmentation. Additionally, data augmentation methods (rotation, affine transformation) were used to improve model learning and decrease overfitting.

The following parameters were used to train the CNN:

Picture Dimension : 96 x 96

Batch size : 32

Rate of Learning : 0.001

Epochs : 40

Loss Function : Cross Entropy Loss

Optimizer : AdamW

In the last few epochs, the model performed at about 98%. The accuracy and loss graphs show consistent learning curves, with accuracy steadily increasing as training goes on.

With most bubbles properly recognized and only a small percentage incorrectly labelled, the confusion matrix demonstrates the CNN model's dependability for automated OMR evaluation.

The accuracy graph, loss graph, and confusion matrix generated throughout the model evaluation procedure are shown below.

5.6.1 Accuracy Analysis

The accuracy graph shows the performance improvement of the CNN-based OMR evaluation model during the training process. The model was trained for 40 epochs, and the accuracy gradually increased with each epoch.

The validation accuracy reached approximately 98% during the final epochs, indicating that the model was able to classify OMR bubbles accurately. The graph also shows stable learning behaviour with only minor fluctuations during training.

The high validation accuracy demonstrates that the CNN model successfully learned the important features required for identifying filled, empty, and noise bubbles from OMR sheets. The model was tested using a separate testing dataset to evaluate real-world prediction performance.

The generated accuracy graph confirms that the proposed CNN model is suitable for automated OMR evaluation and examination result processing in the ExamAIR system.

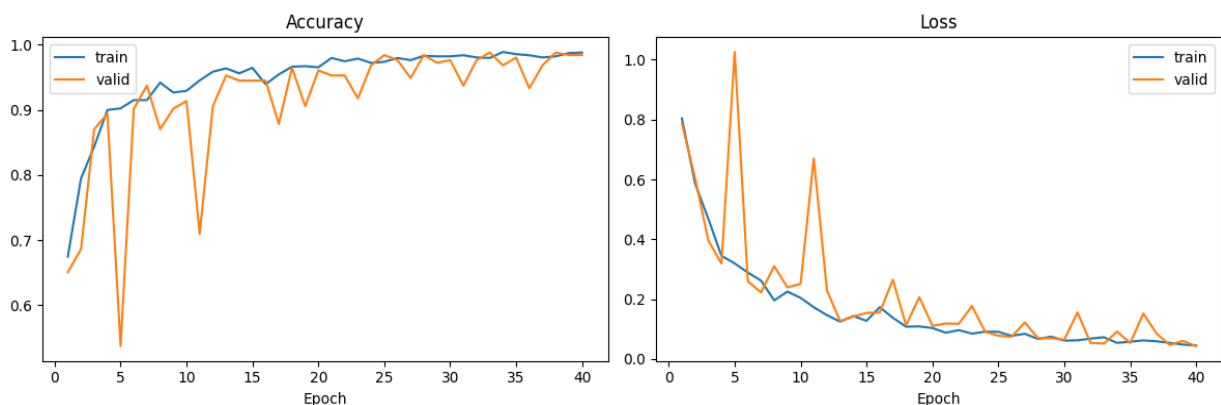


Fig 30 - Accuracy & Loss graph

5.6.2 Confusion Matrix Analysis

A confusion matrix was generated to evaluate the classification performance of the CNN-based OMR evaluation model. The model classified OMR bubbles into three categories:

- Empty
- Filled
- Noise

The confusion matrix shows that:

- All 90 empty bubbles were correctly classified.
- 88 filled bubbles were correctly detected, with only a small number of misclassifications.
- 74 noisy bubbles were correctly classified with very few prediction errors.

The model achieved an overall testing accuracy of approximately 98.8%, confirming the effectiveness of the CNN architecture and OpenCV preprocessing methods.

The low number of classification errors demonstrates that the model can reliably detect marked and unmarked OMR bubbles, which is essential for accurate automated examination evaluation.

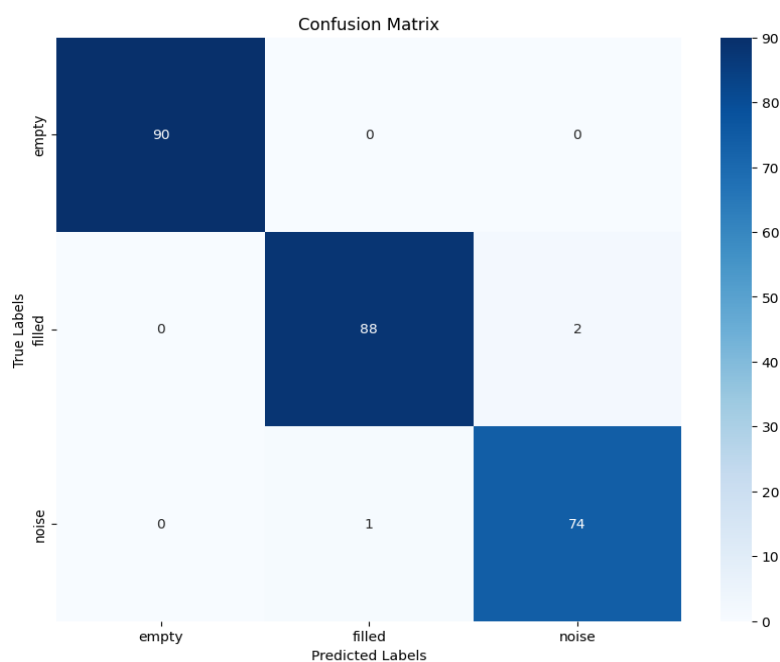


Fig 31 - confusion matrix

5.6.3 Loss Graph Analysis:

The loss graph shown in Figure 30 represents the variation in training loss and validation loss during the CNN model training process. Initially, the loss values were high, but they gradually decreased as the number of training epochs increased.

The reduction in both training and validation loss indicates that the model successfully learned the important features required for OMR bubble classification. Minor fluctuations were observed in validation loss during some epochs; however, the overall downward trend confirms stable learning behaviour and proper model optimization.

The final low loss values demonstrate that the CNN model achieved good convergence and improved prediction capability. The loss stabilization during the final epochs further confirms that the model can perform reliable real-time OMR evaluation with high accuracy.

The generated loss graph validates the effectiveness of the proposed CNN-based approach used in the ExamAIR system for automated examination evaluation and result processing.

6. CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

A modern and effective digital system for managing and grading OMR-based tests is called ExamAIR. Exam administration, OMR sheet processing, evaluator assignment, result production, and rank list publication are all made easier by the system's integration of technologies including Django, Flutter, REST APIs, OpenCV, and CNN-based deep learning models.

The project successfully reduces manual labour, lowers evaluation errors, and improves the speed and accuracy of processing exam results. With special features including automated OMR evaluation, secure role-based access, mobile-based OMR scanning, and centralized examination management, the solution is dependable, user-friendly, and automated.

The OMR assessment module, which is based on CNN and OpenCV, greatly improves the accuracy of responses. By automatically identifying filled bubbles and comparing them with submitted answer sheets, the CNN and OpenCV-based OMR evaluation module greatly improves the accuracy of answer detection. In order to ensure efficiency and transparency in the results release, the system also manages the automatic computation of rank lists and scores.

In conclusion, ExamAIR is an intelligent and scalable examination management system that is appropriate for educational institutions and testing agencies that administer extensive OMR-based exams. The project demonstrates how deep learning techniques and modern software technologies can improve the accuracy, dependability, and efficiency of digital inspection systems.

6.2 FUTURE ENHANCEMENT

The ExamAIR system can be further improved by adding following additional features and advanced technologies to increase efficiency, accuracy, and usability.

1. Intelligent Result Analysis

Machine learning and AI-based techniques can be incorporated to analyze student performance, generate examination statistics, and identify performance patterns automatically.

2. QR Code and Barcode Support

OMR sheets can include QR codes or barcodes for faster identification of candidates and secure handling of examination data.

3. Multi-Language Functionality

Support for multiple languages can be added to make the system more user-friendly and accessible for users from different regions.

4. Cloud-Based Data Storage

Cloud integration can be implemented for secure storage, backup, and centralized management of examination records and scanned OMR sheets.

5. Online Examination Features

Future versions of the system may support online examinations, digital answer submission, and automated result generation along with OMR-based evaluation.

6. Visual Analytics Dashboard

An advanced dashboard can be introduced to display graphical reports related to student performance, examination statistics, evaluator activities, and result summaries.

7. Enhanced OMR Detection Accuracy

The CNN-based OMR detection model can be improved using larger datasets and advanced training methods to achieve better accuracy for low-quality or improperly marked OMR sheets.

8. Student Notification Services

Email or SMS notification features can be integrated to provide updates regarding examinations, evaluator assignments, and published results.

9. Offline Scanning and Synchronization

Offline OMR scanning and temporary local storage features can be added so that evaluators can continue evaluation even without continuous internet access.

These improvements can help make ExamAIR a more scalable, reliable, and efficient digital examination evaluation platform.

7.APPENDICES

7.1 INTERNSHIP CERTIFICATE



10-06-2026



INTERNSHIP EXPERIENCE CERTIFICATE

This is to certify that Ms. Amrutha Balan (Reg. No. IT24GMCAD21), a student of Master of Computer Applications (MCA), Department of IT, Kannur University, completed a Six-Month Academic Internship at Ai2Gi from 22 December 2025 to 10 June 2026 as part of her final semester project and internship requirements. During the internship period, she actively participated in web application development, machine learning research, deep learning model implementation, and software maintenance activities. She demonstrated strong technical capabilities, analytical thinking, and a proactive approach toward learning and problem-solving.

Technologies Used

- Programming Languages: Python, PHP, JavaScript
- Frameworks: Django, Laravel
- Frontend Technologies: HTML, CSS, JavaScript
- Database: MySQL
- Machine Learning & AI: Machine Learning Models, CNN Models, NLP
- Tools & Platforms: Git, GitHub, Jira, Visual Studio Code, REST API

Key Responsibilities and Contributions

- Developed and maintained web applications using Django and Laravel frameworks.
- Designed and implemented a responsive and user-friendly frontend using HTML, CSS, and JavaScript.
- Worked on backend development, REST API integration, and database management using MySQL.
- Built, trained, tested, and evaluated machine learning and deep learning models.
- Developed and implemented CNN models for image processing, classification, and prediction tasks.
- Applied NLP techniques for text analysis, crime data analysis, and prediction systems.
- Created and managed datasets for machine learning model training, validation, and evaluation.
- Utilised Git for source code management, version control, and collaborative software development.
- Used Jira for project management, task tracking, issue reporting, and team collaboration.
- Performed software testing, debugging, troubleshooting, and performance optimisation activities.
- Contributed to feature enhancements, application maintenance, and continuous improvement of systems.

Throughout her internship, Ms Amrutha demonstrated strong analytical and problem-solving skills, maintained a professional attitude, and collaborated effectively with the development team. We appreciate her efforts and commitment during her internship tenure and wish her success in her future academic and professional endeavors.

Sincerely,

Sudhina Sathyan M

CFO & HR - Ai2Gi, Rishanka Group Pvt Ltd

CIN: U73100KL2026PTC100787

+91 83300 64438
+91 99805 87600

www.ai2gi.com
info@ai2gi.com

D25, MiZone
Mangattuparamba, Kannur



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